

Regression Discontinuity Design

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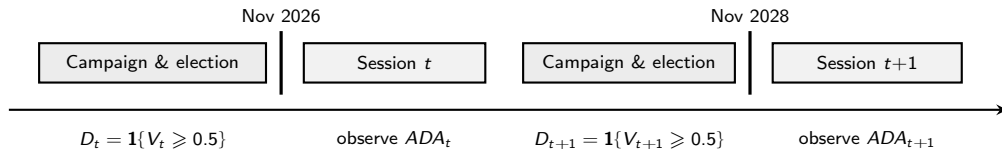
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Institutional Background

US House of Representatives elections

- Elections every **2 years**. 435 congressional districts, each elects one representative.
- **Roll-call votes**: recorded on legislation; summarized into voting indices (e.g. adjusted ADA score).
- **Incumbency advantage**: holding the seat in a district tends to increase win probability next cycle.
- Timing convention: treat t as an **electoral cycle** (includes campaign & election + Congressional session)



- **Running variable**: Democratic two-party vote share in election t (district $\times$ election cycle).
- **Treatment**: whether Democrats win at t : $D_t = \mathbf{1}\{V_t \geq 0.5\}$.

Two Competing Theories

Reference: Lee, D. S., Moretti, E., & Butler, M. J. (2004). Do voters affect or elect policies? Evidence from the U. S. house. *The Quarterly Journal of Economics*, 119(3), 807–859.

Theory A: voters *affect* policies (Downsian)

- Competition for votes induces politicians to move toward the center.
- Exogenous changes in **electoral strength** will change intended policy, and thus roll-call behavior. Stronger electoral strength $\Rightarrow$ weaker incentive to moderate
- $\Rightarrow$ a close win at t also changes incentives (intended policy), so among winners at $t+1$ we should see greater moderation from once-challengers

Theory B: voters *elect* policies

- Politicians implement their own ideal policies if elected.
- Exogenous changes in **electoral strength** do **not** change roll-call behavior.
- $\Rightarrow$ a close Democratic win at t raises $\mathbb{P}(D_{t+1} = 1)$, and ADA_{t+1} shifts **only** because of the party label at $t+1$

Estimating Framework

$$ADA_t = \text{constant} + \pi_0 P_t^* + \pi_1 D_t + \varepsilon_t,$$

$$ADA_{t+1} = \text{constant} + \pi_0 P_{t+1}^* + \pi_1 D_{t+1} + \varepsilon_{t+1}.$$

- P_t^* : Democrats' underlying **electoral strength**; **not directly observed**.
- π_1 : **party effect** on voting records (Dem vs Rep difference holding other factors constant).
- π_0 : **sensitivity of intended policy to electoral strength**.
 - *elect-only / full divergence*: $\pi_0 = 0$.
 - *affect / partial convergence*: $\pi_0 \neq 0$
- ε_t : heterogeneity in ideology ("bliss points") across districts, etc.

Estimating Framework

$$\begin{aligned}ADA_t &= \text{constant} + \pi_0 P_t^* + \pi_1 D_t + \varepsilon_t, \\ADA_{t+1} &= \text{constant} + \pi_0 P_{t+1}^* + \pi_1 D_{t+1} + \varepsilon_{t+1}.\end{aligned}$$

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Key Assumptions

- D_t is quasi-random around cutoff ($V_t \in (0.48, 0.52)$) $\Rightarrow D_t \perp \varepsilon_t, P_t^*$
- "Bliss points" are exogenous (not influenced by who won the previous election) $\Rightarrow D_t \perp \varepsilon_{t+1}$

Estimating Framework

$$\begin{aligned}\gamma &\equiv \mathbb{E}[ADA_{t+1} \mid D_t = 1] - \mathbb{E}[ADA_{t+1} \mid D_t = 0] \\ &= \underbrace{\pi_0 \left(\mathbb{E}[P_{t+1}^* \mid D_t = 1] - \mathbb{E}[P_{t+1}^* \mid D_t = 0] \right)}_{\text{"Affect" component}} \\ &\quad + \underbrace{\pi_1 \left(\mathbb{E}[D_{t+1} \mid D_t = 1] - \mathbb{E}[D_{t+1} \mid D_t = 0] \right)}_{\text{"Elect" component}},\end{aligned}$$

where

$$\pi_1 = \mathbb{E}[ADA_t \mid D_t = 1] - \mathbb{E}[ADA_t \mid D_t = 0]$$

Data

Core sample

- House elections linked across adjacent cycles, spanning roughly 1946–1995 (district $\times$ cycle observations).
- Some adjacent-year pairs are excluded when districts cannot be linked due to redistricting.

Key variables

- Democratic two-party vote share V_t and win indicator D_t .
- Roll-call voting outcomes:
 - Preferred: **adjusted ADA score** for each Representative and Congress/session (higher = more liberal).
- For V.B covariate balance: district demographics (income, education, race, voting-age/eligibility proxies, population, region), drawn from decennial Census (why clustering at district-decade is natural).

$$\text{Total Effect } \gamma = \mathbb{E}[ADA_{t+1} \mid D_t = 1] - \mathbb{E}[ADA_{t+1} \mid D_t = 0]$$

A close win at t has a large impact on *future* voting records ADA_{t+1} .

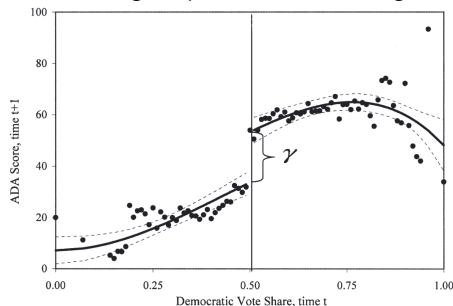


FIGURE I

Total Effect of Initial Win on Future ADA Scores: γ

This figure plots ADA scores after the election at time $t + 1$ against the Democrat vote share, time t . Each circle is the average ADA score within 0.01 intervals of the Democrat vote share. Solid lines are fitted values from fourth-order polynomial regressions on either side of the discontinuity. Dotted lines are pointwise 95 percent confidence intervals. The discontinuity gap estimates

$$\gamma = \underbrace{\pi_0(P_{t+1}^{*D} - P_{t+1}^{*R})}_{\text{"Affect"}} + \underbrace{\pi_1(P_{t+1}^{*D} - P_{t+1}^{*R})}_{\text{"Elect"}}$$

Party Score Gap $\pi_1 = E[ADA_t | D_t = 1] - E[ADA_t | D_t = 0]$

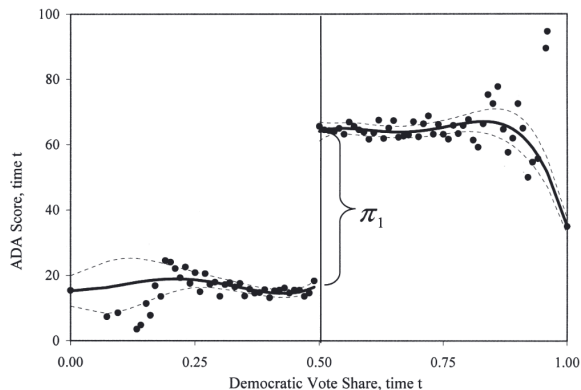


FIGURE IIa
Effect of Party Affiliation: π_1

Incumbency Advantage $E[D_{t+1} | D_t = 1] - E[D_{t+1} | D_t = 0]$

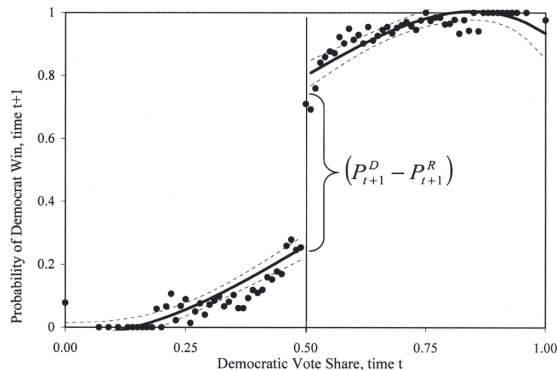


FIGURE IIb

Effect of Initial Win on Winning Next Election: $(P_{t+1}^D - P_{t+1}^R)$

Top panel plots ADA scores after the election at time t against the Democrat vote share, time t . Bottom panel plots probability of Democrat victory at $t + 1$ against Democrat vote share, time t . See caption of Figure III for more details.

Back-of-envelope Decomposition

“Elect” component explains essentially **all** of γ ; “Affect” component is statistically and economically near zero.

TABLE I
RESULTS BASED ON ADA SCORES—CLOSE ELECTIONS SAMPLE

Variable	Total effect			Elect component	Affect component
	γ	π_1	$(P_{t+1}^D - P_{t+1}^R)$	$\pi_1[(P_{t+1}^D - P_{t+1}^R)]$	$\pi_0[P_{t+1}^{*D} - P_{t+1}^{*R}]$
	ADA_{t+1}	ADA_t	DEM_{t+1}	(col. (2))*(col. (3))	(col. (1)) - (col. (4))
	(1)	(2)	(3)	(4)	(5)
Estimated gap	21.2 (1.9)	47.6 (1.3)	0.48 (0.02)	22.84 (2.2)	-1.64 (2.0)

Standard errors are in parentheses. The unit of observation is a district-congressional session. The sample includes only observations where the Democrat vote share at time t is strictly between 48 percent and 52 percent. The estimated gap is the difference in the average of the relevant variable for observations for which the Democrat vote share at time t is strictly between 50 percent and 52 percent and observations for which the Democrat vote share at time t is strictly between 48 percent and 50 percent. Time t and $t + 1$ refer to congressional sessions. ADA_t is the adjusted ADA voting score. Higher ADA scores correspond to more liberal roll-call voting records. Sample size is 915.

Balance Tests

Valid RDD requires: *no systematic discontinuities* in predetermined covariates at cutoff.

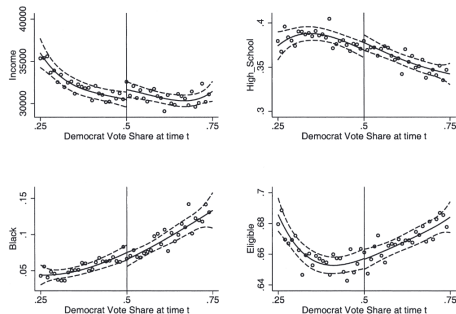


FIGURE III

Similarity of Constituents' Characteristics in Bare Democrat and Republican Districts—Part 1

Panels refer to (from top left to bottom right) the following district characteristics: real income, percentage with high-school degree, percentage black, percentage eligible to vote. Circles represent the average characteristic within intervals of 0.01 in Democrat vote share. The continuous line represents the predicted values from a fourth-order polynomial in vote share fitted separately for points above and below the 50 percent threshold. The dotted line represents the 95 percent confidence interval.

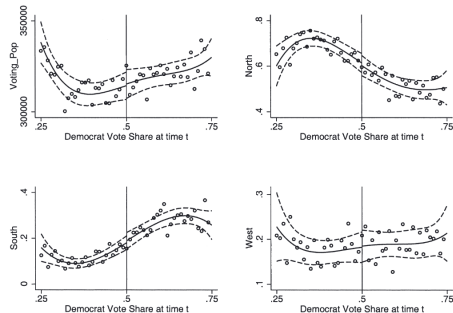


FIGURE IV

Similarity of Constituents' Characteristics in Bare Democrat and Republican Districts—Part 2

Panels refer to (from top left to bottom right) the following district characteristics: voting population, North, South, West. Circles represent the average characteristic within intervals of 0.01 in Democrat vote share. The continuous line represents the predicted values from a fourth-order polynomial in vote share fitted separately for points above and below the 50 percent threshold. The dotted line represents the 95 percent confidence interval.

Placebo Test: Lagged Voting Patterns

Lagged ADA should not jump at 0.5 if assignment near cutoff is quasi-random.

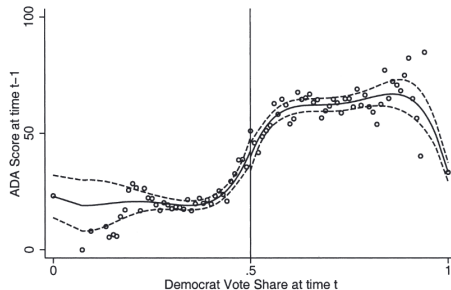


FIGURE V

Specification Test: Similarity of Historical Voting Patterns between Bare Democrat and Republican Districts

The panel plots one time lagged ADA scores against the Democrat vote share. Time t and $t - 1$ refer to congressional sessions. Each point is the average lagged ADA score within intervals of 0.01 in Democrat vote share. The continuous line is from a fourth-order polynomial in vote share fitted separately for points above and below the 50 percent threshold. The dotted line is the 95 percent confidence interval.

Conclusion

- Close elections provide quasi-experimental variation in which party holds the seat going into the next cycle.
- The RD jump in ADA_{t+1} is large, but it is almost entirely accounted for by the “**elect**” channel
- Interpretation: in U.S. House general elections, voters primarily **elect** policies, rather than **affect** candidates' roll-call behavior via competitive moderation.

Review: Modern RDD Practices

- Prefer **local linear** (or local quadratic) with data-driven bandwidth selection over high-order global polynomials.
- Use **robust bias-corrected** inference and show sensitivity to bandwidth choices.
- (i) outcome RD plot, (ii) density/manipulation check, (iii) covariate balance, (iv) placebo outcomes.
- * This paper: 4-th order polynomial, (i), (iii), (iv)

Exercise: Double Machine Learning (Chernozhukov et al., 2018)

Reference: Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., & Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters: Double/debiased machine learning. *The Econometrics Journal*, 21(1), C1–C68.

Consider the partially linear model for a random sample of observations $W_i = (Y_i, D_i, X_i)$:

$$Y_i = D_i\theta_0 + g_0(X_i) + U_i, \quad \mathbb{E}[U_i | X_i, D_i] = 0$$

$$D_i = m_0(X_i) + V_i, \quad \mathbb{E}[V_i | X_i] = 0$$

The goal is to estimate the true scalar structural parameter θ_0 , where g_0 and m_0 are unknown, potentially complex nuisance functions. Assume all variables have bounded second moments and appropriate regularity conditions hold.

Exercise

$$Y_i = D_i\theta_0 + g_0(X_i) + U_i, \quad \mathbb{E}[U_i | X_i, D_i] = 0$$
$$D_i = m_0(X_i) + V_i, \quad \mathbb{E}[V_i | X_i] = 0$$

Q1. Define the score function:

$$\psi(W_i; \theta, g, m) = (Y_i - D_i\theta - g(X_i))(D_i - m(X_i))$$

Show that if the true nuisance functions g_0 and m_0 were known, θ_0 is identified by the moment condition:

$$\mathbb{E}[\psi(W_i; \theta_0, g_0, m_0)] = 0$$

Exercise

$$Y_i = D_i\theta_0 + g_0(X_i) + U_i, \quad \mathbb{E}[U_i | X_i, D_i] = 0$$
$$D_i = m_0(X_i) + V_i, \quad \mathbb{E}[V_i | X_i] = 0$$

Q2. Show that the score $\psi(W_i; \theta, g, m)$ satisfies the Neyman orthogonality condition. Specifically, prove that it is locally insensitive to small perturbations in g and m , such that for any regular directions h_g and h_m :

$$\frac{\partial}{\partial r} \mathbb{E} [\psi(W_i; \theta_0, g_0 + rh_g, m_0 + rh_m)] \Big|_{r=0} = 0$$

Exercise

$$Y_i = D_i\theta_0 + g_0(X_i) + U_i, \quad \mathbb{E}[U_i | X_i, D_i] = 0$$

$$D_i = m_0(X_i) + V_i, \quad \mathbb{E}[V_i | X_i] = 0$$

Q3. Suppose the nuisance functions are estimated on an auxiliary sample, yielding $\hat{g}$ and $\hat{m}$ with convergence rates:

$$\|\hat{g} - g_0\|_{L^2} = O_p(n^{-a}), \quad \|\hat{m} - m_0\|_{L^2} = O_p(n^{-b})$$

The “leading bias” refers to the dominant approximation error when evaluating the expected score with the estimated nuisance functions $(\hat{g}, \hat{m})$ instead of the true functions (g_0, m_0) . i.e.

$$\mathbb{E}[\psi(W_i; \theta_0, \hat{g}, \hat{m}) - \psi(W_i; \theta_0, g_0, m_0)]$$

Show that the leading bias is bounded by $\|\hat{g} - g_0\|_{L^2}\|\hat{m} - m_0\|_{L^2}$. Based on this, what condition on $a + b$ is required for this bias to vanish in the asymptotic distribution of $\sqrt{n}(\hat{\theta} - \theta_0)$?

Exercise

$$Y_i = D_i\theta_0 + g_0(X_i) + U_i, \quad D_i = m_0(X_i) + V_i$$

Q4. The K -fold cross-fitting procedure is explicitly defined by the following steps:

- ① Randomly partition the full dataset of n observations into K roughly equal-sized disjoint subsets (folds), denoted as $I_1, I_2, \dots, I_K$.
- ② For each fold $k \in \{1, \dots, K\}$, use all observations *outside* of fold k (denoted as I_{-k}) to train the machine learning models. This yields the out-of-sample nuisance estimators $\hat{g}_{-k}$ and $\hat{m}_{-k}$.
- ③ Evaluate the moment condition (score) for the observations *inside* fold k ($i \in I_k$) using the estimators $\hat{g}_{-k}$ and $\hat{m}_{-k}$ trained on the auxiliary sample I_{-k} .
- ④ Average the evaluated scores across all K folds to solve for the final parameter estimate $\hat{\theta}$.

Briefly explain why this is necessary when using machine learning algorithms to estimate g_0 and m_0 .

Solution

Q1: Substitute the structural equation $Y_i - D_i\theta_0 - g_0(X_i) = U_i$ and the first-stage equation $D_i - m_0(X_i) = V_i$ into the expectation. By the Law of Iterated Expectations:

$$\mathbb{E}[U_i V_i] = \mathbb{E}[V_i \mathbb{E}[U_i | X_i, D_i]] = 0$$

Solution

Q2: Evaluate the perturbed score:

$$\psi(W_i; \theta_0, g_0 + rh_g, m_0 + rh_m) = (U_i - rh_g(X_i))(V_i - rh_m(X_i))$$

Differentiating with respect to r and evaluating at $r = 0$ yields:

$$-h_g(X_i)V_i - h_m(X_i)U_i$$

Taking the expectation and applying the Law of Iterated Expectations gives zero:

$$\mathbb{E}[-h_g(X_i)V_i] - \mathbb{E}[h_m(X_i)U_i] = -\mathbb{E}[h_g(X_i)\mathbb{E}[V_i | X_i]] - \mathbb{E}[h_m(X_i)\mathbb{E}[U_i | X_i]] = 0$$

(Note that $\mathbb{E}[U_i | X_i] = \mathbb{E}[\mathbb{E}[U_i | X_i, D_i] | X_i] = 0$)

Solution

Q3: Expanding the expected score around the true nuisance functions leaves only the second-order cross-term (first-order terms vanish by Neyman orthogonality proven in part b):

$$\mathbb{E}[\psi(W_i; \theta_0, \hat{g}, \hat{m}) - \psi(W_i; \theta_0, g_0, m_0)] = \mathbb{E}[(\hat{g}(X_i) - g_0(X_i))(\hat{m}(X_i) - m_0(X_i))]$$

By the Cauchy-Schwarz inequality, this term is bounded by:

$$\|\hat{g} - g_0\|_{L^2} \|\hat{m} - m_0\|_{L^2} = O_p(n^{-(a+b)})$$

For the estimator $\sqrt{n}(\hat{\theta} - \theta_0)$ to be asymptotically normal and unbiased, the scaled bias $\sqrt{n}O_p(n^{-(a+b)})$ must vanish. This requires $a + b > 1/2$.

Solution

Q4:

- 1 ML estimators are typically trained to minimize in-sample loss, so $\hat{g}(X_i)$ can be correlated with U_i for observations i used in training - this violates the conditional mean-zero property that the theory requires. Cross-fitting ensures $\hat{g}_{-k}$ and $\hat{m}_{-k}$ are functions of data *independent* of the fold- k observations, restoring the independence structure needed for the CLT argument.
- 2 Unlike a simple sample-split (train on half, evaluate on half), K -fold cross-fitting uses *all* n observations for both estimation and evaluation, preserving efficiency.

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